

Skyworth: Improving the OTT viewing and operation experience with glitchless, easy-to-deploy solutions

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SKYWORTH

By running its OTT in Google Cloud, is able to smooth OTT content to users across regions, save deployment time, quickly expanding overseas business

About Skyworth

Founded in 1988 in China, Skyworth has decades of experience in manufacturing electronic appliances. Skyworth Digital, one of its subsidiaries, offers digital hardware like set-top boxes and smart home devices along with corresponding software solutions. Its OTT solutions are currently run by operators in dozens of countries around the world and have more than 128 million end users.

Industries: Media & Entertainment

higher software
compatibility.

Location: China

Media CDN

Google Cloud results

- Helps reduce 70% of operational workloads with GKE and Cloud Storage
- Eliminates content delivery glitches for OTT system end users through Media CDN
- Lowers operational costs by 15% with custom VMs in Compute Engine

Reduce
solution
deploy
from n
weeks



About WonderCloud

Established in 2021, WonderCloud is a professional cloud computing service company headquartered in Hong Kong. Using industry best practices, WonderCloud helps customers decompose core strategic goals to provide cloud and cloud integrated services, including architecture design, data migration, operation and maintenance hosting, big data and AI platform, cloud management, and security. In May 2022, it obtained Google Cloud's Premier Level Partner Certificate and became a professional partner of Google Cloud.

WonderCloud helps customers
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Google Cloud (<https://cloud.google.com/>)
cloud and cloud integrated services,

Cloud Storage (<https://cloud.google.com/storage/>)

Compute Engine

(<https://cloud.google.com/compute/>)

management and security. In May 2022, it

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

Media CDN (<https://cloud.google.com/media-cdn>)

- Enables
90% of
annual
business
growth in
the OTT
system
market

Over-the-top (OTT) video content has become the main program source on TV in many households around the world. Through one OTT system running on set-top boxes, TV sticks, or smart TVs, users can enjoy easy access to a wide range of video content from broadcast TV shows, streaming media to online social platforms.

To offer quality viewing experiences, operators have to make sure that their software is highly compatible with other applications running on OTT devices, and that the content available meets their local audience's needs. Therefore, adopting a customizable OTT system that comes with matching hardware becomes the go-to solution.



Skyworth product

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—Louis Fang, Overseas
Business R&D Director,
Skyworth

With such challenges in mind, Skyworth (<https://en.skyworthdigital.com/>) is committed to making it easier to provide premium OTT services for operators globally. Founded in 1988 in China, the electronic appliance manufacturer has been developing software solutions for digital hardware, including a modular OTT system that can be easily adapted to different use scenarios, in recent years. This is following its expertise in producing set-top boxes and smart home devices in the past. Its OTT devices and solutions are currently run by operators across the globe, with more than 128 million end users.

"Operators and users in different countries have their own specific needs for the OTT applications and content," explains Louis Fang, Overseas Business R&D Director at Skyworth. "By offering a modular OTT system, we're able to build custom solutions to deliver quality OTT user experiences."

When Skyworth began to expand its OTT business beyond China, it decided to run its OTT system in a public cloud platform that can reduce deployment and operational workloads compared to on-prem infrastructure. The company eventually chose Google Cloud (<https://cloud.google.com/>) because of its extensive global network and high compatibility with Android TV (<https://www.android.com/tv/>) and Google TV (<https://tv.google/>), which are two popular smart TV operating systems worldwide.

"Running our OTT system in a cloud environment can simplify deployment and operational work, which would facilitate our overseas business expansion," notes Fang. "Google Cloud not only provides highly available cloud computing resources, but also industry-leading global connectivity and full integration with smart TV operating systems that can help us optimize the user experience."

Shortening OTT system deployment time from months to weeks

With the immediate technical advice provided by Google Cloud partner WonderCloud, the Skyworth team smoothly migrated the existing OTT systems of its operator customers to Google Cloud, each within one week. For example, when Skyworth needed to adopt the content delivery technology of Google Cloud for the first time in late 2022, an engineer at WonderCloud offered step-by-step guidance on the migration so they could quickly get onboard the new tool.

Now, Skyworth runs the applications of its OTT system that change constantly with containers in Google Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine>) (GKE), and the ones that are more stable with virtual engines (VMs) in Compute Engine

(<https://cloud.google.com/compute>). It also leverages Cloud Storage (<https://cloud.google.com/storage>) to store video files for inter-process communication (IPC) and Media CDN

(<https://cloud.google.com/media-cdn/docs/overview>) to deliver content around the world.

Fang says that the whole cloud infrastructure has significantly shortened the time required to deploy Skyworth's OTT system, because all the computing resources are instantly available, and the containerized architecture of GKE and the content delivery mechanism of Media CDN already optimized for streaming videos help save time required to set up its system in a new environment.

"If we use on-prem hardware to deploy our OTT system, it would have taken around half a year to purchase all the devices and finish the software settings. With the infrastructure of Google Cloud, we can complete the whole deployment process within one month," he adds.

Enhancing service stability for lower operational workloads and better user experience

For Skyworth, another advantage of adopting Google Cloud is the enhanced stability of its services. The

autoscaling feature of GKE ensures that Skyworth's OTT system can smoothly tackle traffic surges, while its node auto-upgrading feature allows all the nodes in the clusters of Skyworth's system to be up-to-date with the latest Kubernetes version.

Furthermore, the various cross-region storage bucket types offered by Cloud Storage enables the Skyworth team to effortlessly make its stored data highly available. Overall, the high stability of the Google Cloud infrastructure and the elimination of on-site technical support resulting from the adoption of cloud have helped reduce 70% of operational workloads.

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*—Louis Fang, Overseas
Business R&D Director,
Skyworth*

On top of that, the higher content delivery performance supported by Media CDN through the widely distributed network nodes of Google Cloud has enabled more stable connectivity that translates into better user experience. Through an internal test, Skyworth found that the traditional content delivery mechanism would lead to high latency or disruptions when users download software updates due to limited bandwidth. With Media CDN, end users of Skyworth's OTT system are able to receive video content and data without glitches.

"With its content delivery mechanism optimized for video content and large data files, Media CDN has helped us ensure glitchless user experience for the end users of our OTT system worldwide," notes Fang.

Facilitating OTT business expansion with lower operational costs and high system integration

Adopting Google Cloud has also helped Skyworth drive up its business performance. With the custom VMs in Compute Engine, the Skyworth team is able to set up servers with the exact amount of CPUs and

memory that it requires, which has helped save 15% of operational costs. As the Google Cloud tools can be fully integrated with Android TV and Google TV, Skyworth's OTT devices and system have received growing popularity since the migration, reaching an annual growth of 90% in the global market share.

"For an OTT software provider like us, leveraging Google Cloud not only means the accessibility of high-performance cloud tools, but also further integration with the wider Google ecosystem. This way, our end users can enjoy a better viewing and user experience on their smart TVs," Fang notes.

Expanding OTT features and automating system management with AI

Next, Skyworth plans to leverage the AI capabilities of Google Cloud to build more innovative features for its OTT service, such as using the generative AI technology to make its system more interactive. The company will also employ Vertex AI (<https://cloud.google.com/vertex-ai>) to automate its backend system management work, which can further reduce operational workloads for its team and its operator customers.

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Fang says, "The infrastructure of Google Cloud has enabled us to provide easy-to-deploy OTT solutions with high connectivity around the world. We're ready to continue expanding our overseas business and develop more innovative features on Google Cloud."